UPSC

Prarambh 2028

→ Advance Class of S&T :

Science & Technology

Lecture No. – 10

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Classification of Orbits (Part 06)

Classification of Satellites:

- Earth Observation Satellite (Remote Sensing Satellite)
  - Active
    - RISAT + HISAR + NISAR + Fireflies
  - Passive
    - CARTOSAT
    - RESOURCESAT
    - OCEANSAT
- Communication Satellite → GSAT9, GSAT24, GSAT7 (7, 7A, 7B, 7R)
- Weather Satellite → INSAT 3D series
  - 3D
  - 3DR
  - 3DS
- Navigation Satellite:
  - Global (MEO) — GPS, Galileo, GLONASS, COMPASS
  - Regional (GSO)
    - NavIC
      - Gen 1
      - Gen 2
    - QZSS
- Astronomical Satellite (Space observation satellite)
  - AstroSAT
  - XPoSat
  - Aditya L1

Navigation Satellite System

Speed = 3L Km/sec

Atomic Clock ⇐ t = 0.0035765 sec

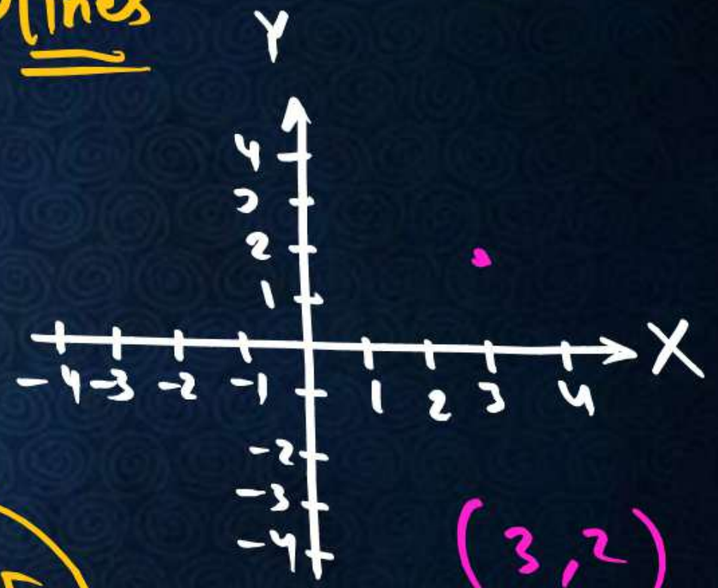
2d = speed × time

$d = \frac{\text{speed} \times \text{time}}{2}$

Longitude → Imaginary lines

Latitude

30°N / 20°E



(3, 2)

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→ At least 3 satellites are required to read the location of a static position.

→ At least 5 satellites are required to read the location of a moving object at any particular time.

Navigation satellites provide autonomous geo-position services. (relative position of a body)

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Payload:

- Transponder (data uplink & downlink will be on same device)
- Atomic Clock: (used for precise time keeping)

Two types of Navigation Satellite

Global Navigation Satellite System [MEO]

→ Min. no. of satellite Required for Global Coverage ⇒ 24

1. GPS [Global Positioning Services] → USA.
2. Galileo → European Union
3. GLONASS → [Global Nav. Sat. System] → Russia
4. Compass → China

Regional Navigation satellite System

→ Orbit ⇒ GSO / GEO

Min. no. of Sat. req ⇒ 5

1. Bei-dou → China
2. QZSS: Quasi Zenith Satellite System → Japan
3. NavIC → Navigation with Indian constellation.
   → Older Name → IRNSS (Indian Regional Navigation Sat. System)

Global Positioning System (GPS) → USA

- launched in 1978
- This is 1st Global Navigation Satellite System
- 24-Satellite Constellation System

→ Originally, it was made for USA Military Purpose only

In 1994, An Air Traffic Control of S. Korea lost its Radar Signal & flight that was ready to land, crashed & there were no survivors left due to this crash.

→ After this incidence & Global Crisis, USA agreed to open its Global GPS System for the rest of the world (for civilian use only)

Post 1994, USA GPS was made to work for 2 different purpose:

Standard Positioning Signal (SPS)

- for civilian purpose
- can be used by all

Precise Positioning Systems (PPS)

- for USA Military use only.
- Signals are highly encrypted

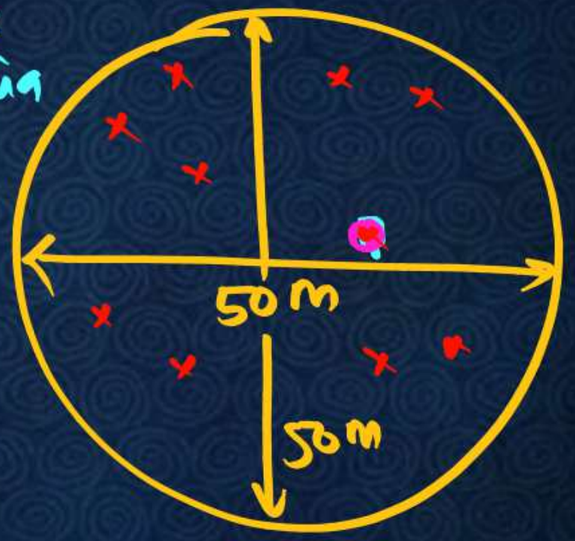
encoded (Chatur Chintu → Psipnc cFn5#J

Kargil War → 1999

↓

GPS Accuracy

- SPS → Accuracy → 50m ⇓ Indian Access
- PPS → 2m



Post Kargil

→ IRNSS: {Constellation of 7 satellite} — 3 Sat ⇒ GEO; 4 Sat ⇒ GSO

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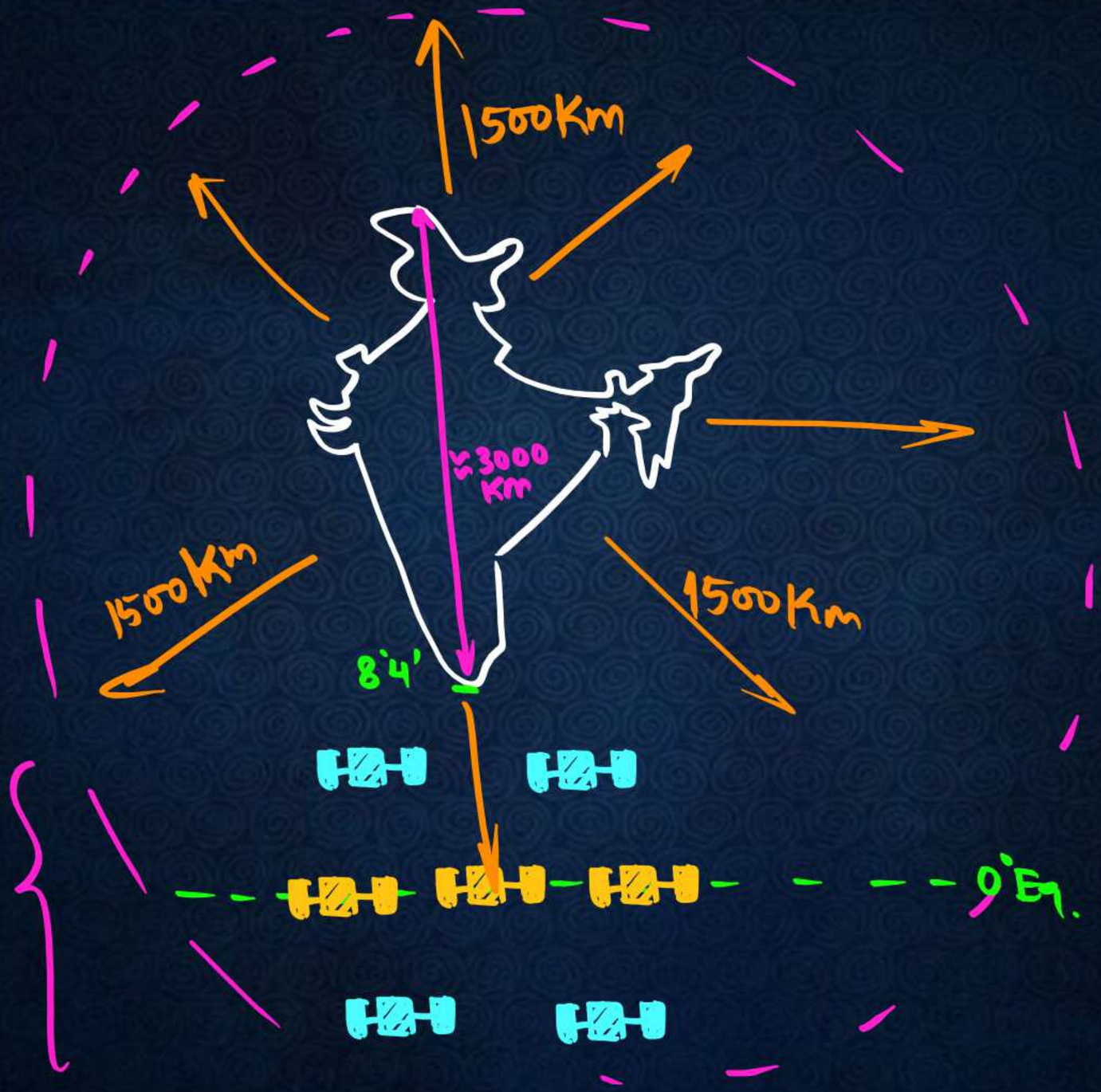

NavIC
- 3sat → GEO
- 4sat → GSO

NavIC:

SBNS: Satellite Based Navigation System

Coverage → India + 1500 Km beyond in every direction

Accuracy:

- Standard Positioning Signal (SPS) → Civilian Purpose
- Reserve signals (Precise positioning signal (PPS)) → only for Military Purpose

→ upto 15m
- from Indian land → upto 10m accuracy
- from water (ocean) surface → upto 20 m accuracy

→ upto 2m

Payload
- Transponder
- Rhubidium Atomic Clock

NavIC Satellites:

1st Generation Satellites: [IRNSS]

- Total Sent ⇒ 9 Satellites
  - Atomic clock of 2 sat. were malfunctioned } × × × } Replace
- Total Sat. working for NavIC (Earlier) → 7
  (Now) → 3
- Light weight Satellites (< 1425 Kg)
  - Sent from → PSLV
- Payload
  - Transponders (DualBand) — L5 Band, S Band
  - Rhubidium Atomic Clock {Imported from European Union}

2nd Generation Navigation Satellites

(NVS 01, 02, 03, 04, 05)

- Sent
  - NVS-01 → successful → 2024.
  - NVS-02 → failed → (Jan 2025) → 100th launched of ISRO
- Launched by GSLV Mark II.
- Heavier satellites
- Payload
  - Transponder — L5 Band, S-Band
  - L1 Band (Newly Added)
  - Indegenously built Rhubidium Atomic Clock } 1st in NVS-01

→ NVS ⇒ long lifespan of ≈ 10 years

- They will be integrated with GAGAN

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Note:

Use of L1 Band: Most common signal used by global Navigation like GPS

- Now 2nd Gen. Nav. Sat will be compatible with GPS & can be integrated with GPS.
- It is used in single frequency-chip → commonly used in mobile phones, wearable devices like smart watch etc.

GAGAN:

[GPS Aided Geo-Augmented Navigation]

SBAS: Satellite Based Augmented System.

It's payload is augmented on Communication Satellites.

→ It will improve the Accuracy of GPS Signals in Indian Air Space & it will help in better air traffic control.

⇒ It is jointly developed by Airport Authority of India (AAI) & ISRO.

→ It provides high precision positioning services across India & other areas.

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Application of GAGAN:

1. Civil Aviation:

- Precise Landing
- 3-dimensional approach to guide on runways.
- Air traffic Management

GPS
- Latitude, Longitude
- Altitude

2. Maritime & Fisheries:

- Mariner Navigation → Enhanced positioning for Boats & ships.
- GEMINI System → GAGAN Enabled Mariner's Instrument for Navigation & Infor^n
  - App → emergency warnings to deep sea fisherman
- Real time disaster Alerts.

3. Railway:

- Unmanned Crossing Safety
- Monitor Train movements.

4. Natural Resources & land management:

- Forest Survey
- Wildlife Monitoring.

5. Road & Transport Management

- NH → Modern management system
- Intelligent Traffic Management.

6. Disaster Management

Thank You !